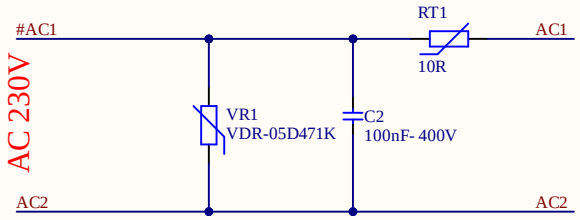
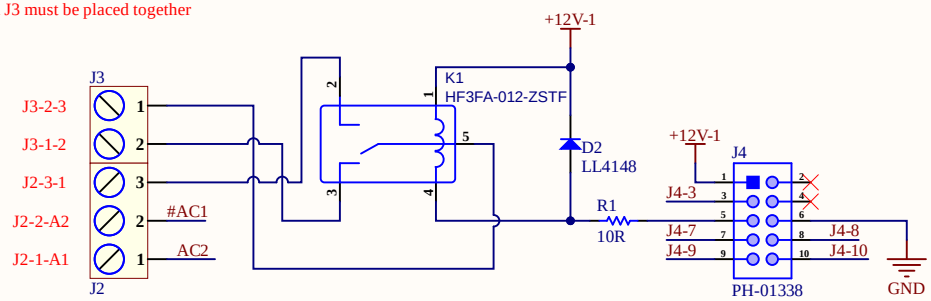
SSR-05D-PIC-1-1.6 (PCB 1)



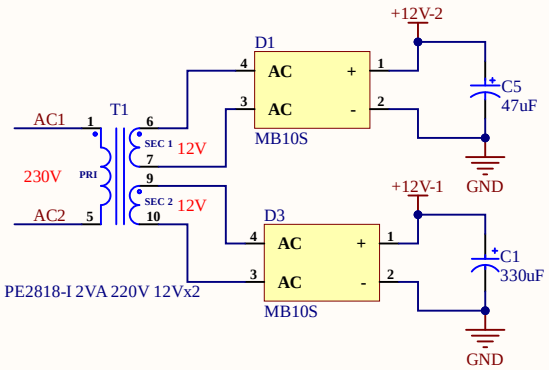
Relay Circuit

J2 and J3 must be placed together

3 = Output
2 = 220V AC (Line)
1 = NC
A1 = 220V AC (Line)
A2 = 220V AC (Neutral)

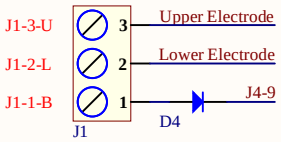


AC (230V) to DC (12V)

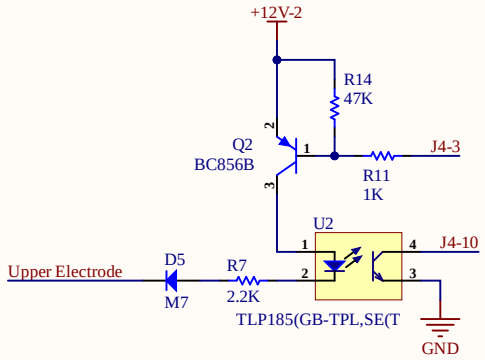


Screw Terminal for Connecting Electrodes

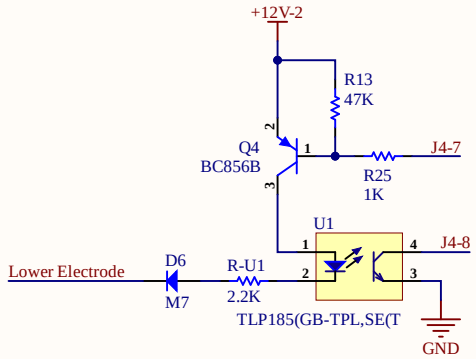
U = Upper Electrode
L = Low Electrode
B = Bottom Electrode



Upper Electrode Control Circuit



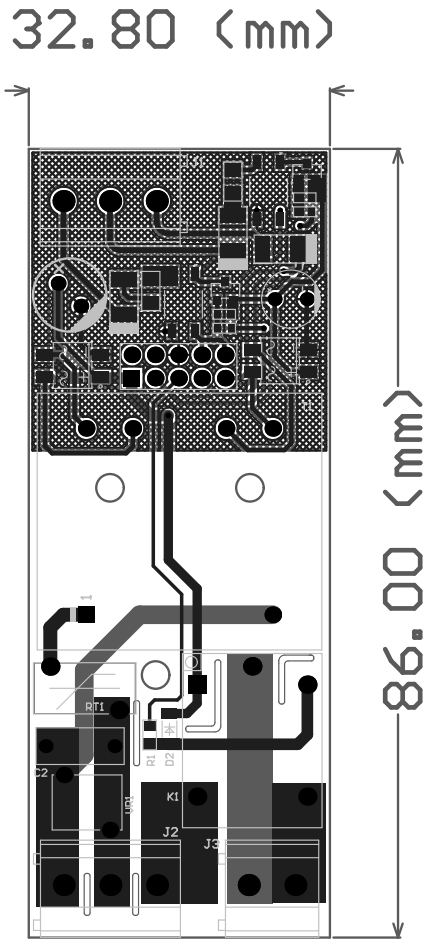
Lower Electrode Control Circuit



eSpark Consulting Group

Title: LLR-05U (Main-Board-Schematic).SchDo	
File: LLR-05U (Main-Board).PrjPcb	
Date: 04/07/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 1

Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.076mm	3.5	
1	Top Layer		0.254mm		
	Dielectric 1	FR-4	0.940mm	4.8	
2	Bottom Layer		0.254mm		
	Bottom Solder	Solder Resist	0.076mm	3.5	
	Bottom Overlay				
Total board thickness:			1.600mm		



Comment	Description	Designator	Footprint	LibRef	Quantity
330uF	330uF 35V ±20% 860mA@100kHz Plugin,D8xL14mm Aluminum Electrolytic Capacitors - Leaded ROHS	C1	RC035M331LO8*14TH-2A1E	RC035M331LO8*14TH-2A1E	1
100nF- 400V	5% 400V 100nF PET Plugin,P=7.5mm Film Capacitors ROHS	C2	82CL0114	82CL0114	1
47uF	47uF 50V ±20% 180mA@100kHz Plugin,D6.3xL12mm Aluminum Electrolytic Capacitors - Leaded ROHS	C5	Plugin,D6.3xL12mm	03EC0571	1
MB10S	35A 1.1V@400mA 1A 1kV MBS Bridge Rectifiers ROHS	D1, D3	MBS10S	MB10S	2
LL4148	75V 1V@10mA 4ns 200mA LL-34 Switching Diode ROHS	D2	LL-34	LL4148	1
M7	1A 1.1V@1A 1kV DO-214AC Diodes - General Purpose ROHS	D4, D5, D6	SMA(DO-214AC)	M7	3
KF129-5.08-3P	1x3P 24A - 40℃~+105℃ 250V Green 12~24 Straight 5.08mm 2.5 1 3 Plugin,P=5.08mm Screw terminal ROHS	J1, J2	KF129-5.08-3P	KF129-5.08-3P	2
KF129-5.08-2P	1x2P 24A - 40℃~+105℃ 250V Green 12~24 Straight 5.08mm 2.5 1 2 Plugin,P=5.08mm Screw terminal ROHS	J3	KF129-5.08-2P	KF129-5.08-2P	1
PH-01338	3A Straight Square Pins 10P 3mm - 40℃~+105℃ 3mm 2.54mm 双排 Black 2x5P 2.54mm Plugin,P=2.54mm Pin Headers ROHS	J4	PH-01338	PH-01338	1
HF3FA-012-ZSTF	12V SPDT (1 Form C) Plugin,15.2x19mm Power Relays ROHS	K1	HF3FA-012-ZSTF	HF3FA-012-ZSTF	1
BC856B	65V 200mW 220@2mA,5V 100mA PNP SOT-23 Bipolar Transistors - BJT ROHS	Q2, Q4	SOT-23	BC856B	2
10R	125mW Thick Film Resistors 150V ±1% ±200ppm/℃ 10Ω 0805 Chip Resistor - Surface Mount ROHS	R1	RES-0805	0805W8F100JT5E	1
2.2K	250mW Thick Film Resistors 200V ±100ppm/℃ ±1% 2.2kΩ 1206 Chip Resistor - Surface Mount ROHS	R7, R-U1	Res-1206	1206W4F2201T5E	2
1K	RES SMD 1K OHM 0.1% 1/10W 0603	R11, R25	Res-0603	RT0603BRD071KL	2
47K	RES SMD 47K OHM 0.1% 1/10W 0603	R13, R14	Res-0603	RT0603BRD0747KL	2
10R	10Ω 2A 2700K Plugin,P=7.5mm NTC Thermistors ROHS	RT1	10D-9	10D-9	1
PE2818-I 2VA 220V 12Vx2	Power 2VA 220V / 6Vx2/7.5Vx2/9Vx2/12Vx2/15Vx2/18Vx2/24Vx2 Encapsulated For PCB Mounted Isolation Transformer	T1	PE2818-I	PE2818-I 2VA 220V 12Vx2	1
TLP185(GB-TPL,SE(T	80V 3.75kV 50mA 300mV@2.4mA,8mA 1.5V 1.25V DC SOP-4-2.54mm Transistor Output Optocoupler ROHS	U1, U2	SOP-4_2.54	TLP185(GB-TPL,SE(T	2
VDR-05D471K	810V 385V 300V 423V~517V Plugin Varistors ROHS	VR1	VDR-05D471K	VDR-05D471K	1